

$$p(x) = x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n$$

$$\xi = \xi(\underbrace{a_1, \dots, a_n}_a) \text{ rădăcină simplă}$$

$$\Rightarrow p(\xi(a)) = 0 \quad (*) \quad \text{și} \quad p'(\xi(a)) \neq 0.$$

Fix $j \in \{1, \dots, n\}$.

Derivăm parțial (*) în rap. cu variabila a_j

$$\Rightarrow \frac{\partial p(\xi(a))}{\partial a_j} = 0$$

$$\Rightarrow \frac{\partial}{\partial a_j} (\xi^n(a) + a_1 \xi^{n-1}(a) + \dots + a_{n-1} \xi(a) + a_n) = 0$$

$$n \cdot \xi^{n-1}(a) \cdot \frac{\partial \xi(a)}{\partial a_j} + a_1 (n-1) \xi^{n-2}(a) \frac{\partial \xi(a)}{\partial a_j} + \dots + (a_j \cdot (n-j) \xi^{n-j-1}(a) \frac{\partial \xi(a)}{\partial a_j} + \xi^n(a)) + \dots + a_{n-1} \frac{\partial \xi(a)}{\partial a_j} = 0$$

$$\Rightarrow p'(\xi(a)) \cdot \frac{\partial \xi(a)}{\partial a_j} + \xi^{n-j}(a) = 0$$

$\vdots$

$$(cond_1 \xi)(a) = \dots$$